

Getein1600 Communication Protocol

Packet Protocol

Baud rate: 9600

Data bits: 8

Stop bit: 1

Parity bit: None (Only for serial port communication)

The data of network protocol packet and serial port packet are the same.

Note: 1) Each data frame consists of STX, CMD, Length, CheckSum and Block, it will be received and displayed in hex.

2) The test results must be compared with “Min” value and “Max” value before display. If the test value is out the range of “Min” and “Max” value, it would be displayed as < Min or > Max, as shown in the following table.

Test result	Measure Range	Display
0 mg/L	Min=0.5 Max=10.0	< 0.5 mg/L
20 mg/L	Min=0.5 Max=10.0	> 10.0 mg/L

1. Data Frame Format

1) Format: STX CMD Len CheckSum Block

2) Introduction:

STX (0x02)	Frame Head	1 byte
CMD (0x01)	Command Mode	1 byte
Len	Length	2 byte
CheckSum	Check	1 byte
Block	Data Packet	N byte

3) Data packet format:

| Barcode | No. | ID | Item | Name | Unit | Value | Min | Max | Ref | Sample | Mode | Datetime | Channel | Position |

Data packet introduction:

Barcode	Barcode, no barcode means blank
No.	Number (The number of test result increased automatically, no modification)
ID	ID number (Users can input number manually and also can modify the number; If no input, it would increase automatically since the number of last test)
Item	Test item number
Name	The name of test item
Unit	The unit of test item
Value	The value of test item
Min	The minimum value of test item
Max	The maximum value of test item
Ref	Reference value
Sample	Sample mode ('1' stands for serum/plasma; '2' stands for whole blood; '3' stands for urine; '4' stands for fingertip blood)
Datetime	The date and time of the test (Format: yyyy-mm-dd HH:MM:SS)
Channel	The detection channel of sample

Position	The location of sample in sample frame
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Note: Different contents separated by “|”, different items in a content separated by “^”, for example | CK-MB^cTnI^Myo | .

2. Heartbeat Packet Format

1) Format: STX CMD Len CheckSum Block

Note: Heartbeat packet only used for serial port connection.

2) Introduction:

STX (0x02)	Frame Head	1 byte
CMD (0x02)	Command Mode	1 byte
Len	Length	2 byte
CheckSum	Check	1 byte
Block (0x00)	Data	1 byte

Note: LIS sends heartbeat to Getein1600, such as 0x02 0x02 0x01 0x00 0x03 0x00.

And Getein1600 answers LIS, such as 0x02 0x02 0x01 0x00 0x04 0x01.